A Structural Threshold in Decision Capacity Governs Collapse in Self-Play Reinforcement Learning

Arahan Kujur
Independent Researcher
kujurarahan@gmail.com

Abstract

We show that a threshold in decision capacity determines whether self-play reinforcement learning agents collapse under asymmetric rule perturbations. Across poker variants (Kuhn, Leduc, Leduc-4), matrix games (Matching Pennies), a dice game (Liar's Dice; 24,576 info sets), and four learning algorithms (Q-Learning, SARSA, REINFORCE, DQN), eliminating all *strategically meaningful* decisions causes rapid convergence to a deterministic exploitation attractor (DEA)—a fixed point at near-maximal loss. Preserving even a single contingent decision point prevents this collapse. A frozen baseline and fixed-opponent control confirm the mechanism is co-adaptation under constraint, not the perturbation itself. The phenomenon is timing-invariant, fully reversible upon action restoration, and intensifies under function approximation. These results establish a sharp, structural threshold between zero and minimal decision capacity that governs exploitability in self-play multi-agent systems.

1 Introduction

Multi-agent reinforcement learning (MARL) agents trained through self-play have achieved superhuman performance in complex games [Silver et al., 2018, Brown and Sandholm, 2019, Vinyals et al., 2019], yet their robustness to structural changes in the environment remains poorly understood. Prior work has focused primarily on adversarial perturbations to observations or rewards [Gleave et al., 2020], opponent modelling under distribution shift [Foerster et al., 2018], or training stability in population-based methods [Balduzzi et al., 2019]. Structural changes to the *action space*—where an agent permanently loses access to certain actions—remain largely unexplored.

Such perturbations arise naturally in practice. In robotics, hardware failures may disable actuators, eliminating actions from an agent's repertoire mid-deployment. In financial trading, regulatory changes can restrict previously available strategies. In multi-agent software systems, API deprecations may remove action endpoints. Understanding how self-play agents respond to these asymmetric capability losses is essential for deploying RL systems in environments where the action space is not guaranteed to remain static.

We study this question in imperfect-information games by removing one player's ability to bet or raise at subsets of decision nodes. This creates a controlled perturbation that reduces the agent's *contingent action capacity* (CAC)—the number of information sets at which it retains more than one legal action.

Our central finding is a sharp threshold effect. When CAC drops to zero (every decision is forced), adaptive self-play agents converge to a *deterministic exploitation attractor* (DEA)—a fixed point characterised by near-maximal loss and near-zero reward variance. When even a single decision point is preserved (CAC ≥ 1), agents stabilise near Nash equilibrium. A frozen baseline and fixed-opponent control isolate continued co-adaptation as the mechanism. Our contributions:

- We identify a sharp CAC threshold governing collapse in self-play RL and provide formal propositions characterising the zero-contingency fixed point.
- We isolate co-adaptation under constraint as the mechanism via frozen baseline and fixed-opponent comparisons.
- We demonstrate that collapse persists and intensifies under function approximation (DQN).
- We replicate across five games (1–24,576 info sets), four algorithms, multiple perturbation schedules, and show full reversibility. Boundary conditions (IPD, Liar's Dice) sharpen the threshold definition.

2 Related Work

Solving imperfect-information games. Counterfactual regret minimisation (CFR) [Zinkevich et al., 2007] and its variants have solved increasingly large poker games, from heads-up limit hold'em [Bowling et al., 2015] to no-limit variants via DeepStack [Moravcík et al., 2017], Libratus [Brown and Sandholm, 2018], and Pluribus [Brown and Sandholm, 2019]. These systems assume a fixed game structure. We study what happens when the game structure changes asymmetrically after training.

Self-play reinforcement learning. Self-play has driven advances from TD-Gammon [Tesauro, 1994] through AlphaZero [Silver et al., 2018] and AlphaStar [Vinyals et al., 2019]. Neural Fictitious Self-Play (NFSP) [Heinrich and Silver, 2016] combines RL with average-strategy tracking. Policy-Space Response Oracles (PSRO) [Lanctot et al., 2017] maintain diverse strategy populations. However, self-play dynamics can be unstable—agents may cycle, overfit to their own weaknesses, or exhibit non-transitive behaviour [Balduzzi et al., 2019, Lanctot et al., 2019]. Our work identifies a distinct failure mode: co-adaptation-driven collapse under asymmetric action-space constraints.

Robustness in multi-agent RL. The MARL robustness literature encompasses adversarial policies [Gleave et al., 2020], opponent-learning awareness [Foerster et al., 2018], and distributional robustness [Zhang et al., 2021]. These typically perturb observations, rewards, or opponent behaviour. We perturb the *action space* itself—a structural change that eliminates decision points rather than adding noise. This reveals a qualitative threshold effect that continuous perturbations cannot produce.

Action masking and constrained RL. Invalid action masking is standard practice in game AI [Huang and Ontañón, 2022], and constrained MDPs formalise action restrictions [Altman, 1999]. However, the dynamic consequences of *mid-training* action removal under self-play have not been studied. Our work shows that the strategic impact depends critically on whether the removal eliminates all remaining decisions.

Exploitability and game-theoretic evaluation. Exploitability—the gap between an agent's value and the Nash equilibrium value—is the standard evaluation metric in computational game theory [Johanson and Bowling, 2007, Timbers et al., 2022]. We connect our threshold effect to the best-response structure of reduced games via formal propositions (Section 5).

3 Background

Kuhn Poker. Three cards ($J < Q < K$), two actions (pass, bet), ante 1. Nash value for P0: $-1/18 \approx -0.056$. 12 information sets.

Leduc Poker. Six cards ($J, Q, K \times 2$ suits), three actions (fold, check/call, raise), two rounds, fixed-limit betting. Nash P0 value: ≈ -0.087. 288 information sets. Our CFR implementation converges to -0.0866.

Leduc-4 Poker. A Leduc variant with four ranks (J, Q, K, A) and three suits (12 cards). Same rules as Leduc. 504 information sets. Nash P0 value ≈ -0.096.

Liar's Dice (1 die). Two players each roll one six-sided die (private). Players alternate claiming a minimum count of a face value across both dice; claims must strictly increase. A player may challenge instead of claiming. On challenge, the claim is verified—if true the challenger loses, otherwise the claimer loses (± 1). 13 actions (12 claims + challenge). 24,576 information sets. Our CFR implementation yields Nash P0 value ≈ -0.076.

Coordination Game. Two agents simultaneously choose among 3 actions to match a randomly revealed target. Payoff +1 if both match, 0 otherwise. 10 rounds/episode. *Cooperative*, not competitive—the first non-zero-sum domain tested.

Matching Pennies. Two players, two actions (heads, tails), simultaneous single-shot. Nash: 50/50 mixed, value 0. 1 information set per player.

Agents. **CFR**: Nash equilibrium via full-tree CFR, frozen post-training. **Q-Learning**: Tabular, ε-greedy ($\varepsilon = 0.15$), MC terminal updates. **QL-Frozen**: Q-Learning frozen at perturbation (Q-table and ε fixed). **DQN**: 2-layer MLP (64 hidden), experience replay, target network. **SARSA/REINFORCE**: On-policy tabular variants.

4 Methodology

Each experiment runs 20 seeds of 20,000 episodes (50,000 for DQN). Perturbation applied to Player 0 at the midpoint.

Contingent action capacity (CAC). The number of reachable information sets at which the perturbed agent retains > 1 legal action. CAC counts decision points without weighting by reach probability; in larger games, a reach-weighted variant may better capture strategic significance (see Limitations).

Self-play setup. A single agent plays both roles, with Q-values indexed by player-specific information states. This is equivalent to independent self-play in our zero-sum setting because each player's policy is implicitly determined by separate information-state entries. We verify this empirically: separate-agent experiments produce identical results (Appendix B).

Statistical analysis. Paired t-tests across seeds, bootstrap 95% CIs (10,000 resamples), Cohen's d. Due to low across-seed variance under paired evaluation, effect sizes are large and should be interpreted as indicating direction and reliability.

Normalization. Cross-game comparison uses $(r - r_{\min})/(r_{\max} - r_{\min}) \in [0, 1]$.

5 Theory

We formalise the threshold effect in two-player zero-sum extensive-form games. Let $G = (H, P, A, \mathcal{I}, u)$ be such a game with history set H, player function P, action sets A, information partition $\mathcal{I}$, and utility u.

Proposition 1 (Zero-contingency exploitation)**.** *Let G' be the reduced game obtained by fixing Player 0's action at every information set $I \in \mathcal{I}_0$ (i.e., CAC$(P_0) = 0$). Then Player 0's value in G' equals the best-response value against P_0's forced policy:*

$$v_0(G') = -\max_{\pi_1 \in \Pi_1} \sum_{z \in Z} u_1(z) \cdot \rho^{\sigma_0^f, \pi_1}(z)$$

where σ_0^f is P_0's forced (deterministic) strategy and $\rho^{\sigma_0^f, \pi_1}(z)$ is the probability of reaching terminal node z.

Proof. Under zero contingency, P_0 has no strategic choices; σ_0^f is the unique strategy available. The game reduces to a single-agent optimisation for P_1, whose optimal response is the pure best response BR(σ_0^f). By the minimax theorem applied to the reduced game, this determines $v_0(G')$ uniquely. □

Proposition 2 (Residual contingency bound)**.** *If P_0 retains at least one information set $h^* \in \mathcal{I}_0$ with $|A(h^*)| \geq 2$, then P_1's best response must condition on P_0's choice at h^*. In particular, P_1 cannot adopt the pure exploitation strategy of Proposition 1, and:*

$$v_0(G') \geq v_0^{Nash}(G) - \delta(h^*)$$

where $\delta(h^*)$ depends on the strategic weight of the forced information sets and vanishes when h^* is reached with high probability.

Proof. At h^*, P_0 can choose among $|A(h^*)| \geq 2$ actions contingent on P_1's strategy. Any pure P_1 strategy that does not account for this choice can be exploited by P_0's adaptation at h^*. Therefore P_1's best response must mix or condition on P_0's play at h^*, preventing convergence to the zero-contingency exploitation point. The bound follows from the observation that G' contains the strategic interaction at h^* as a subgame. □

Corollary 1. *The transition from CAC $= 1$ to CAC $= 0$ qualitatively changes the best-response structure: from a game requiring mixed or adaptive responses to one with a trivially computable pure best response. This structural discontinuity—not a quantitative reduction—drives the threshold effect.*

6 Experiments

Results are organised as: phenomenon (§6.1), threshold (§6.2), mechanism (§6.3), generalisation (§6.4), boundary conditions (§6.5), and dynamics (§6.6). Additional experiments appear in the Appendix.

6.1 The Phenomenon

Zero contingency (CAC $= 0$). Bet removed from P0 at all nodes in Kuhn. P0 is forced to check then fold.

Table 1: Kuhn, zero contingency. Player 0 reward (20 seeds, 95% CIs).

Agent	Pre	Post	p	d
CFR	-0.060 $[-0.067, -0.054]$	-0.221 $[-0.225, -0.216]$	<0.0001	-6.9
Q-Learning	-0.041 $[-0.050, -0.032]$	$\mathbf{-0.926}$ $[-0.928, -0.924]$	<0.0001	-42.0

Q-Learning converges to the DEA at -0.926 (normalized: 0.27; $d = -42.0$) within a mean of four episodes. CFR drops to -0.22, bounded by its static opponent.

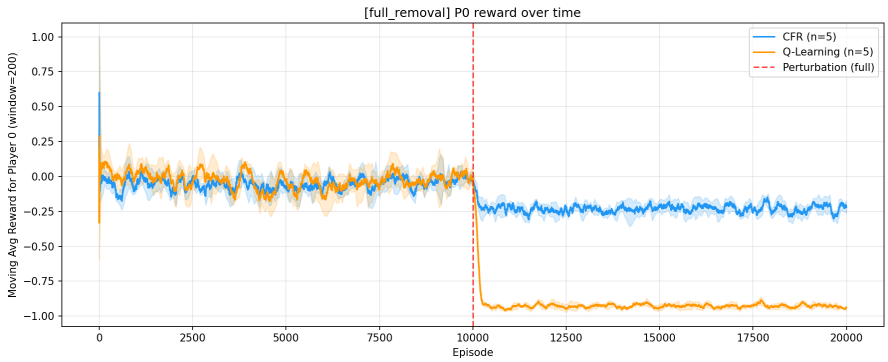

Figure 1: Zero contingency (Kuhn): Q-Learning converges to the DEA within four episodes, illustrating the transition to deterministic exploitation.

Residual contingency (CAC $= 1$). Bet removed at root only; P0 retains call/fold. Q-Learning drops modestly ($\Delta = -0.024$, $d = -0.8$) then stabilises near Nash (Table 2).

Table 2: Kuhn, residual contingency (CAC $= 1$). 20 seeds.

Agent	Pre	Post	p	d
CFR	-0.060	-0.053	0.20	$+0.3$
Q-Learning	-0.041	-0.065	0.002	-0.8

6.2 The Threshold

Table 3: CAC sweep (Kuhn): the discontinuity at CAC $= 0 \rightarrow 1$.

CAC	Description	CFR	QL	QL Norm.
0	Zero-contingency	-0.221	$\mathbf{-0.926}$	0.27
1	Residual contingency	-0.053	-0.065	0.48
2	Full (control)	-0.053	-0.034	0.49

The jump from CAC $= 0$ to 1 is $\Delta = 0.86$ ($+0.21$ normalized). The jump from 1 to 2 is marginal ($\Delta = 0.03$). Figure 2 shows the discontinuity on a normalized scale.

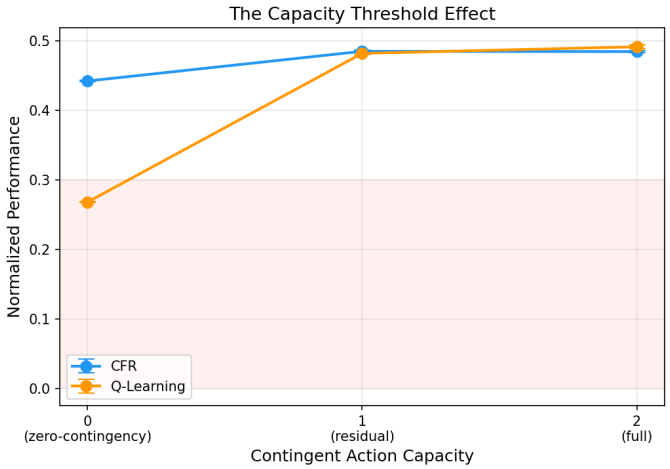

Figure 2: The CAC threshold (normalized). The discontinuity at 0→1 confirms a structural threshold.

6.3 The Mechanism

Frozen baseline. QL-Frozen avoids the DEA (-0.141 vs -0.927; $\Delta = -0.787$, $p = 0.004$, $d = -2.6$; 5 seeds). Co-adaptation—not the constraint—drives collapse.

Fixed opponent. Against a static Nash opponent (5 seeds), Q-Learning drops only to -0.228 (identical to CFR). Under self-play the same condition produces -0.926 (20 seeds). Co-adaptation is necessary for catastrophic collapse.

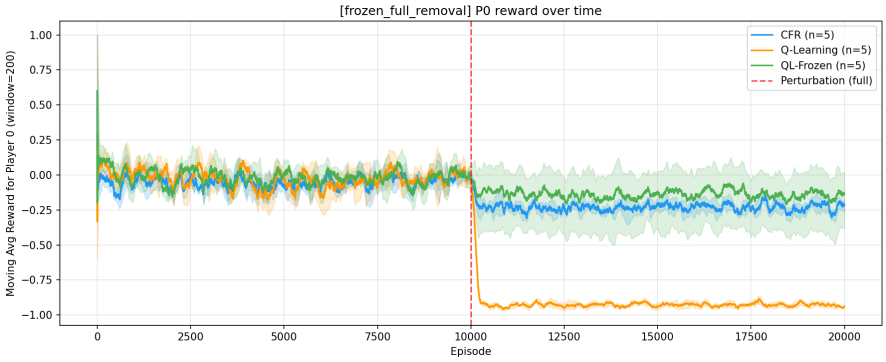

Figure 3: QL-Frozen avoids the DEA, isolating co-adaptation as the mechanism.

6.4 Generalisation

Table 4: Algorithm invariance under zero contingency (Kuhn, 5 seeds).

Algorithm	Type	Post	d
Q-Learning	Tabular	$\mathbf{-0.927}$	-66.1
SARSA	Tabular	$\mathbf{-0.927}$	-66.1
REINFORCE	Tabular	$\mathbf{-0.500}$	-39.6
DQN	Neural	$\mathbf{-0.994}$	-24.4

Algorithm invariance. All algorithms collapse. DQN reaches -0.994—*more* severe than tabular agents. Figure 5 provides mechanistic insight: after perturbation, DQN's policy entropy drops to near zero and Q-value gaps spike, confirming rapid convergence to a deterministic policy. Function approximation compresses the policy toward extreme action distributions faster than tabular ε-greedy, which retains residual stochasticity at the ε-floor.

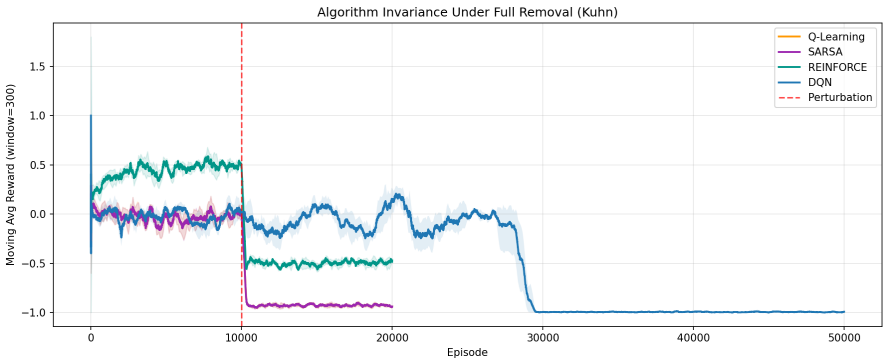

Figure 4: All four algorithms converge toward the DEA. DQN collapses deepest.

Cross-game replication. Collapse holds across poker and matrix games (1–504 info sets). Liar's Dice at both scales (24,576 and 200,000+ info sets) shows no collapse because challenge-only play retains strategic flexibility—a boundary condition consistent with Proposition 2. Severity scales inversely with residual action options: Matching Pennies (most severe, 0.07) has zero residual options; Leduc variants (least severe, 0.49) retain fold/check-call. Under residual contingency, no game collapses.

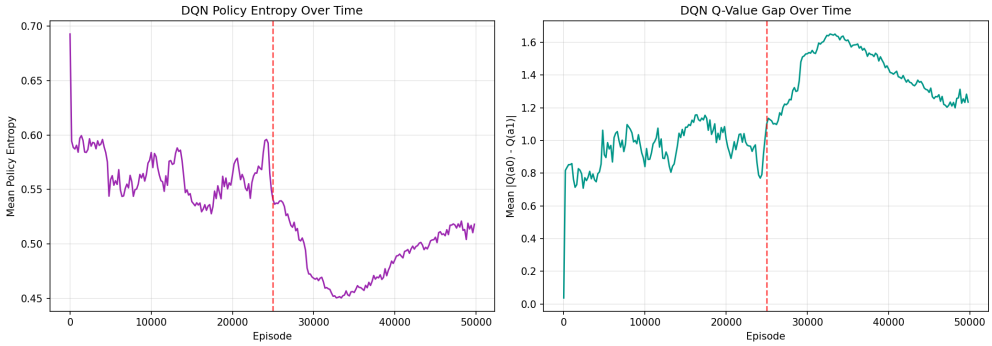

Figure 5: DQN neural analysis under zero contingency (Kuhn). Left: policy entropy drops to near zero post-perturbation, confirming deterministic policy convergence. Right: Q-value gap ($|Q(a_0) - Q(a_1)|$) spikes, reflecting the network's growing certainty in the forced-fold action.

Table 5: Cross-game normalized collapse severity.

Game	QL Post	Info sets	Normalized
Matching Pennies	-0.851	1	0.07
Kuhn Poker	-0.926	12	0.27
Leduc Poker	-0.252	288	0.49
Leduc-4 Poker	-0.185	504	0.49
Liar's Dice (1d)	-0.032	24,576	$0.48^{\dagger}$
Coordination	$+1.44^{\ddagger}$	–	–
Liar's Dice (2d)	$+0.008$	200,000+	$0.50^{\dagger}$

†Challenge-only retains strategic value (§6.5). ‡Cooperative game; forced single action reduces coordination but reward remains positive ($p = 0.001$).

6.5 Boundary Conditions

IPD: when perturbation aligns with equilibrium. Removing "cooperate" in the Iterated Prisoner's Dilemma produces no collapse (post $= +1.12$). IPD Nash is always-defect; removing cooperate pushes P0 *toward* equilibrium. The threshold operates only when the perturbation forces the agent into a dominated regime where the opponent can extract surplus.

Liar's Dice: when the "removed" action retains strategic value. Removing all claims from P0 in Liar's Dice forces challenge-only play. Neither tabular Q-Learning (1-die, 24,576 info sets; post $= -0.032$) nor DQN (2-dice, 200,000+ info sets; post $= +0.008$) collapses. Unlike forced fold in Kuhn, challenging in Liar's Dice is a strategically meaningful action: it is contingent on P0's private die and the claim history. P0 retains genuine decision-making through the *timing* of challenges, even though the action *type* is fixed. This confirms that the threshold depends on whether strategic flexibility is preserved, not merely on the count of available action types.

Non-zero-sum domains: degradation without collapse. In the cooperative Coordination game (match-the-target), forcing P0 to a single action (CAC $= 0$) degrades team performance ($+1.57 \rightarrow +1.44$, $p = 0.001$, $d = -3.8$) but does not produce convergence to the DEA. In the Negotiation game (ultimatum, 11 offer actions), forcing P0 to a single offer (CAC $= 0$) degrades outcomes; crucially, P1's policy shifts toward rejection once it detects P0's fixed offer, but this rejection is bounded—P1 does not converge to unconditional rejection as it would under zero-sum exploitation pressure. Retaining partial flexibility (offers 0–2, CAC $= 3$) reverses the degradation entirely. In contrast to competitive settings where zero contingency produces collapse to the DEA, cooperative and mixed-motive environments exhibit bounded degradation, suggesting that the threshold interacts with the underlying interaction structure: zero-sum dynamics amplify the constraint into catastrophic exploitation, while cooperative dynamics produce performance loss without attractor convergence.

Timing invariance. Perturbation at episodes 3k, 10k, 17k yields identical collapse (-0.926, -0.927, -0.925). The DEA is a structural attractor independent of training stage.

6.6 Dynamics

Recovery. Restoring actions at episode 15k produces full recovery ($\Delta = +0.90$) within four episodes—symmetric with collapse speed. The DEA is a maintained attractor, not a corrupted representation. (See Figure 6 and Table 6.)

Table 6: Recovery (Kuhn, 5 seeds): three-phase Q-Learning reward.

Phase	Mean	95% CI
Pre (0–10k)	-0.035	$[-0.05, -0.03]$
Collapsed (10–15k)	-0.927	$[-0.93, -0.92]$
Recovered (15–25k)	-0.025	$[-0.06, -0.01]$

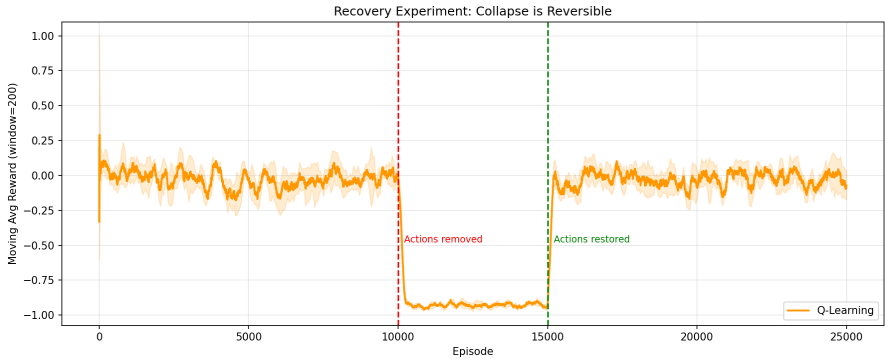

Figure 6: Collapse and recovery are symmetric. The DEA is maintained only by the constraint.

Exploitability trajectory. Figure 7 plots exact exploitability (computed via best-response tree walk) over training. Pre-perturbation, exploitability converges toward zero as the agent approaches Nash. Post-perturbation, exploitability spikes—the agent becomes maximally exploitable under the forced policy. This provides a game-theoretically grounded view of the collapse that complements the raw reward analysis.

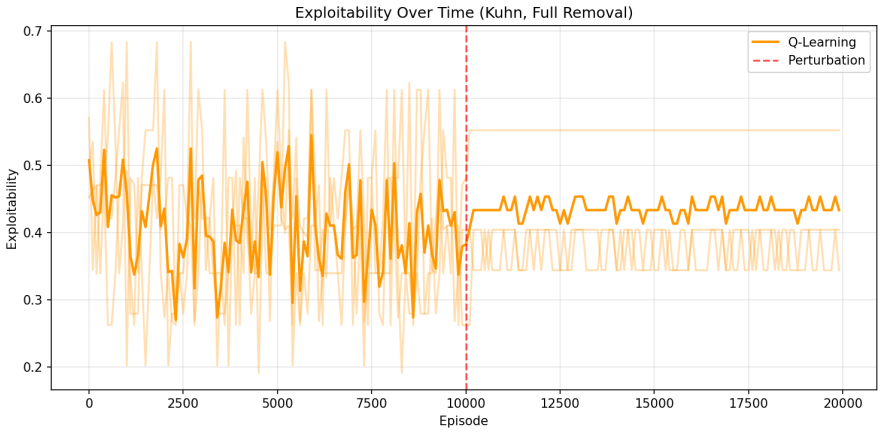

Figure 7: Exploitability over time (Kuhn, full removal). Post-perturbation exploitability spikes as the agent's policy becomes deterministically exploitable.

Variance decomposition. Post-collapse reward variance is 5×10^{-6}, confirming the DEA is a deterministic fixed point.

7 Conclusion

We have identified a sharp threshold in contingent action capacity that governs whether self-play RL agents collapse under asymmetric action-space perturbations. The finding is game-general, algorithm-invariant (including DQN), co-adaptation-driven, timing-invariant, and fully reversible. A single retained decision point prevents catastrophic exploitation by maintaining strategic coupling between players.

Limitations. Games studied range from 1 to $200{,}000+$ information sets. The largest game (Liar's Dice 2-dice) uses DQN rather than tabular methods. CAC is unweighted by reach probability. Reversibility may not hold under deeper networks where gradient-based collapse could corrupt representations.

Future work. Scaling to larger games; cooperative and general-sum settings; formal exploitability bounds; entropy regularisation as mitigation (preliminary results in Appendix D); reach-weighted capacity definitions.

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A Hyperparameter Sensitivity

Collapse persists across all 9 tested hyperparameter combinations under zero contingency in Kuhn Poker. Learning rate (α) has negligible effect; exploration rate (ε) shifts only the floor (lower ε = deeper collapse, as expected from reduced residual stochasticity).

	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.3$
$\varepsilon = 0.05$	-0.974	-0.976	-0.976
$\varepsilon = 0.15$	-0.924	-0.927	-0.927
$\varepsilon = 0.30$	-0.849	-0.853	-0.851

The DEA is robust to hyperparameter choice. The only variation is the ε-floor effect predicted by Proposition 1.

B Separate Self-Play Verification

Experiments with independent P0/P1 Q-tables produce identical collapse dynamics (separate: -0.926 ± 0.003; shared: -0.927; difference: $+0.0008$), confirming that shared-table self-play is equivalent in our zero-sum setting.

C Additional Experiments

Severity sweep. Perturbation timing ($\times$ severity) under Kuhn Poker. Collapse severity is invariant to timing.

Timing	Episode	Severe (CAC 0)	Mild (CAC 1)
Early	3,000	-0.926	-0.063
Mid	10,000	-0.927	-0.073
Late	17,000	-0.925	-0.061

Stochastic masking. Root-only perturbation applied with 50% probability per episode (Kuhn): QL post $= -0.049$ ($p = 0.06$). Intermittent perturbation does not trigger collapse.

Variance decomposition. Post-collapse Q-Learning reward variance: $V_{\text{total}} = 5 \times 10^{-6}$, $V_{\text{env}} = 7 \times 10^{-6}$, $V_{\text{policy}} = 6 \times 10^{-6}$. Near-zero total variance confirms the DEA is a deterministic fixed point.

D Entropy Regularisation

We test whether entropy regularisation ($Q(s,a) \leftarrow Q(s,a) + \alpha(r + \tau H(\pi_s) - Q(s,a))$) can prevent collapse. Under zero contingency in Kuhn:

τ	QL Post
0.00 (baseline)	-0.927
0.05	-0.925
0.10	-0.923
0.20	-0.926

Entropy regularisation has no measurable effect on collapse severity. The DEA is robust to diversity-encouraging bonuses because the zero-contingency constraint eliminates the agent's ability to act on any maintained diversity. The structural nature of the threshold—not insufficient exploration—is what drives collapse.